



MASENO UNIVERSITY
UNIVERSITY EXAMINATIONS 2017/2018

**THIRD YEAR SECOND SEMESTER EXAMINATIONS FOR THE
DEGREE OF BACHELOR OF SCIENCE IN ACTUARIAL SCIENCE
WITH INFORMATION TECHNOLOGY**

MAIN CAMPUS

MAC 300: FINANCIAL MATHEMATICS II

Date: 6th July, 2018

Time: 3.30- 6.30 pm

INSTRUCTIONS

- Answer question ONE and any other TWO questions
- Observe further instructions on the answer booklet



QUESTION 1 [COMPULSORY] [30 Marks]

(a) Differentiate between a par yield and a gross redemption yield of a fixed income security. [6 Marks]

(b) Show that

$$F_{t,r} = \frac{1}{r} \log_e \frac{P_t}{P_{t+r}}$$

where $F_{t,r}$ is a continuous time forward rate of duration r from time t and P_t is the price of a zero-coupon bond at time t . [6 Marks]

(c) An insurance company has liabilities of Kshs 6 million due in exactly 8 year's time and a further Kshs 11 million due in exactly 15 year's time. The assets held by the insurance company consist of:

- a 5-year zero-coupon bond of nominal amount Kshs 5.5088 million;
- and
- a 20-year zero-coupon bond of nominal amount Kshs 13.7969 million.

The current rate of interest is 8% per annum effective at all durations. Show that the first two conditions of Redington's theory for immunisation against small changes in the rate of interest are satisfied. [6 Marks]

(d) The random variable $(1 + i)$ has a log-normal distribution with parameters μ and σ^2 .

(i) Assuming the standard results for the log-normal distribution, show that

$$E(i) = \exp(\mu^2 + \frac{1}{2}\sigma^2) - 1$$

and

$$\text{Var}(i) = \exp(2\mu + \sigma^2)(\exp(\sigma^2) - 1)$$

[6 Marks]

(ii) Let $E(i) = j$ and $\text{Var}(i) = s^2$, show that

$$\sigma^2 = \log_e \left[1 + \left(\frac{s}{1+j} \right)^2 \right]$$

and

$$\mu = \frac{1+j}{\sqrt{\left[1 + \left(\frac{s}{1+j} \right)^2 \right]}}$$

[6 Marks]

QUESTION 2 [20 Marks]

Let f_t denote the one-year effective forward rate of interest over the year from time t to $(t + 1)$. Let it be the t -year effective spot rate over the period 0 to t . The annual effective gross redemption yield from an n -year bond which pays coupons of 5% annually in arrear is given by:

$$g_n = 0.07 + 0.001n$$

for $n = 1, 2$ and 3

Each bond is redeemed at par and is exactly one year from the next coupon payment.

- (a) Calculate i_1 , i_2 and i_3 as percentages to three decimal places. [12 Marks]
- (b) Calculate f_0 , f_1 and f_2 as percentages to three decimal places.

[8 Marks]

QUESTION 3 [20 Marks]

Three bonds, each paying annual coupons in arrear of 4% and redeemable at par, reach their redemption dates in exactly one, two and three years time, respectively. The price of each bond is 96 per 100 nominal.

- (a) Calculate the gross redemption yield of the three-year bond. [6 Marks]
- (b) Calculate, showing all workings, the one-year and two-year spot rates of interest implied by the information given. [6 Marks]
- (c) Calculate the forward rate of interest applicable over the second year.

[4 Marks]

(d) Explain whether the three-year spot rate will be higher than or lower than the three-year gross redemption yield.

[4 Marks]

QUESTION 4 [20 Marks]

An investment fund has liabilities of Kshs.11 million due in 7 years time and Kshs.8.084 million in 11 years time. The manager of the fund will meet the liabilities by investing in zero-coupon bonds. The manager is able to buy zero-coupon bonds for whatever term is required and there are adequate funds at the managers disposal.

(a) Explain whether it is possible for the manager to immunise the fund against small changes in the rate of interest by purchasing a single zero-coupon bond.

[6 Marks]

(b) The manager decides to purchase two zero-coupon bonds, one paying Kshs.15.363 million in 7.5 years time and the other paying Kshs.3.787 million in 14.25 years time. The current interest rate is 5.5% per annum effective. Determine whether the investment fund satisfies the necessary conditions to be immunised against small changes in the rate of interest.

[14 Marks]

QUESTION 5[20 Marks]

An insurance company has just written single premium contracts that require it to make payments to policyholders of Kshs.10,000,000 in five years time. The total single premiums paid by policyholders amounted to Kshs.8,000,000. The insurance company is to invest the premiums in assets that have an uncertain return. The return from these assets in year t , i_t , is independent of the returns in all previous years with a mean value of 5.5% per annum effective and a standard deviation of 4% per annum effective. $(1 + i_t)$ is lognormally distributed.

(a) Calculate, deriving all necessary formulae, the mean and standard deviation of the accumulation of the premiums over the five-year period.

[14 Marks]

(b) A director of the company is concerned about the possibility of a considerable loss from the investment in the assets suggested in part (a). Instead, the director suggests investing in fixed interest securities with a guaranteed return of 4% per annum effective. Set out the arguments for and against the directors position.

[6 Marks]



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MAIN CAMPUS

MAC 302: METHODS OF ACTUARIAL INVESTIGATIONS II

Date: 2nd July, 2018

Time: 3.30- 6.30 pm

INSTRUCTIONS

- Answer question ONE and any other TWO questions.



QUESTION ONE

- (a) Explain what is meant by the principle of correspondence. [2 marks]
- (b) Describe briefly some validity checks that can be carried out to ascertain the reliability of the data to be used for a mortality investigation. [4 marks]
- (c) You have been given the information from the mortality experience of a large life office covering the years 2007 and 2008. Your information includes the number of people aged x nearest birthday at 1 January 2007, 2008 and 2009, and the number of deaths during 2007 and 2008 aged x nearest birthday. Derive the crude initial rates of mortality q_x . State the assumptions that you had to make. [3 marks]
- (d) A mortality investigation is being conducted between the dates 1 January 2008 and 31 December 2011. The data from four lives under consideration was as follows:
- | Life | Date of birth | Date of entry | Date of exit | Mode of exit |
|--------------|---------------|---------------|--------------|---------------|
| <i>Pele</i> | 11.11.77 | 24.3.06 | 29.12.11 | Death |
| <i>Johan</i> | 1.9.84 | 30.8.09 | — | Did not leave |
| <i>Gray</i> | 10.2.83 | 10.10.07 | 21.6.08 | Surrender |
| <i>Diego</i> | 8.2.86 | 10.8.09 | — | Did not leave |
- i. Assuming that the day of entry, but not the day of exit, counts in the exposed to risk, calculate the number of days of exposure contributed to the central exposed to risk by each life at each age. [12 marks]
- ii. State what modifications, if any, you would need to make if you were determining the initial exposed to risk and not the central exposed to risk. [3 marks]
- (e) A graduation of a set of mortality rates from age 25 to age 64 has 15 positive individual standardised deviations, which occurred in 8 groups.
- Carry out two tests to check the suitability of this graduation. [6 marks]

QUESTION TWO

- (a) i. If mortality follows Gompertz Law such that $\mu_x = 0.00003 \times 1.1^x$, calculate the values of μ_{80} , μ_{81} , μ_{82} , μ_{83} (to 7 decimal places) and the first, second and third differences derived from these quantities. [3 marks]
- ii. Recalculate differences using the observed rates $\hat{\mu}_{80} = \mu_{80} - 0.01$, $\hat{\mu}_{81} = \mu_{81}$, $\hat{\mu}_{82} = \mu_{82}$, $\hat{\mu}_{83} = \mu_{83}$. [2 marks]
- iii. Use the values calculated in (i) and (ii) to explain why the test for smoothness usually requires the third differences of the graduated rates to be small, rather than the differences of a lower or a higher order. [3 marks]
- (b) i. For a mortality investigation you have been given data relating to the initial exposed to risk and number of deaths for ages 60 to 94. Describe how you would carry out a graphical graduation and show how you would derive approximate 95% confidence intervals for the age specific mortality rates. [7 marks]
- ii. Describe how you would test your graduated rates for smoothness. [1 mark]
- iii. In the investigation the following data have been collected.

Age last birthday	Initial exposed to risk	Number of deaths
60 – 64	20500	374
65 – 69	27800	892
70 – 74	30100	1475
75 – 79	26700	2158
80 – 84	17700	2251
85 – 89	9200	1780
90 – 94	2300	630

State, giving reasons, whether the graphical method is appropriate for the graduation of this experience, or whether a different method would be more suitable.

[4 marks]

QUESTION THREE

- (a) A mortality investigation was held between 1 January 2011 and 1 January 2013. The following information was collected. The figures in the table below are the numbers of lives on each census date with the specified age labels.

Age last birthday	Date		
	1.1.11	1.1.12	1.1.13
48	3486	3384	3420
49	3450	3507	3435
50	3510	3595	3540

During the investigation there were 42 deaths at age 49 nearest birthday. Estimate μ_{49} stating any assumptions that you make.

[7 marks]

- (b) i. State the principle of correspondence as it applies to the estimation of mortality rates. [1 mark]
- ii. Explain why it might be difficult to ensure the principle of correspondence is adhered to, and give a specific example of an investigation where this may be the case. [2 marks]
- iii. An actuary was asked to investigate the mortality of lives in a particular geographical area. Data are available of the population of this area, classified by age last birthday, on 1 January in each year. Data on the number of deaths in this area in each calendar year, classified by age nearest birthday at death, are also available.
Derive a formula which would allow the actuary to estimate the force of mortality at age $x + f$, μ_{x+f} , in a particular calendar year, in terms of the available data, and derive a value for f . [6 marks]
- iv. List four factors other than geographical location which a government statistical office might use to subdivide data for national mortality analysis. [2 marks]
- (c) At a particular age there are 922 deaths, compared to 950 expected. Calculate the approximate individual standardised deviation for that age. [2 marks]

QUESTION FOUR

- (a) i. Describe three shortcomings of the χ^2 test for comparing crude estimates of mortality with a standard table and why they may occur. [3 marks]

The following table gives an extract of data from a mortality investigation conducted in the rural highlands of a developed country. The raw data have been graduated by reference to a standard mortality table of assured lives.

Age	Expected deaths	Observed deaths	z_x
60	36.15	35	-0.191
61	28.92	24	-0.915
62	31.34	27	-0.775
63	38.01	35	-0.488
64	26.88	32	0.988
65	37.59	36	-0.259
66	33.85	34	0.026
67	26.66	32	1.034
68	22.37	26	0.767
69	18.69	33	3.310
70	18.24	22	0.880

- ii. For each of the three shortcomings you described in (i):
 A. name a test that would detect that shortcoming.
 B. carry out the test on the data above. [12 marks]
- iii. Comment on your results from (ii). [3 marks]
- (b) List four factors other than age and smoker status by which life insurance mortality statistics are often subdivided. [2 marks]

QUESTION FIVE

- (a) The mortality experience of a large company pension scheme is to be tested to see if the experience of males aged 65-72 years is consistent with a standard table. The results were collated by the firm conducting the analysis on a computer spreadsheet, with positive and negative standardised deviations being distinguished only by being in a different colored font. Unfortunately the results have been supplied to the company in the form of a printout produced on a black-and-white printer from which it is not possible to tell the signs of the deviations. The values of the standardised deviations shown are as follows

0.052, 0.967, 2.528, 0.328, 1.234, 0.25, 1.023, 0.756

Carry out two tests which could be conducted from the information given. [10 marks]

- (b) The actuary to a large pension scheme carried out an investigation of the mortality of the schemes pensioners over the two years from 1 January 2005 to 1 January 2007.

- i. List the data required by the actuary for an exact calculation of the central exposed to risk for lives aged x . [2 marks]

The following is an extract from the data collected by the actuary.

<i>Age nearest birthday</i>	<i>No. of pensioners at</i>			<i>Deaths during</i>	
	1.1.2005	1.1.2006	1.1.2007	2005	2006
63	1248	1312	1290	10	6
64	1465	1386	1405	13	15
65	1678	1720	1622	16	23
66	1719	1642	1667	22	19
67	1686	1695	1601	19	25

- ii. Derive an expression that could be used to estimate the central exposed to risk using the available data. State any assumptions you make. [4 marks]
- iii. Use the data to estimate μ_{64} . State any further assumptions that you make. [2 marks]

- (c) A graduation of a set of crude mortality rates is tested for goodness-of-fit using a chi-squared test.

Discuss the factors to be considered in determining the number of degrees of freedom to use for the test statistic. [2 marks]



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MAIN CAMPUS

MAC 304: ACTUARIAL LIFE CONTINGENCIES

Date: 4th July, 2018

Time: 12.00 - 3.00 pm

INSTRUCTIONS

- Attempt question ONE and any other TWO questions
- Show all your workings
- Observe neatness and clarity in your workings



QUESTION ONE

- (a) Explain the meaning of the following concepts
- Insurance policy lapse
 - Select mortality rates
 - Life annuity
 - Surrender value
 - Reversion [5 marks]
- (b) i. 30 percent of those who die between ages 25 and 75 die before age 50. The probability of a person aged 25 dying before age 50 is 0.2. Find the probability that a person age 20 years and three months. [2 marks]
- ii. Prove that for a life aged x , each of the following is true.
- $l_x - l_{x+n} = d_x + d_{x+1} + \cdots + d_{x+n-1}$ [1 mark]
 - ${}_m+n p_x = ({}_m p_x) \cdot ({}_n p_{x+m})$ [1 mark]
- iii. Compute the net single premium for a 100,000 life insurance policy payable at the moment of death, purchased by a life aged 30, if it is assumed that interest is 6 percent and ${}_t p_{30} = (0.98)^t$ for all t . [3 marks]
- (c) i. Show that the force of mortality for two independent lives (x) and (y) is given by
- $$\mu_{xy} = \mu_x + \mu_y$$
- [3 marks]
- ii. Find an expression for each of the following;
- The probability that a 30-year-old dies before age 60. [1 mark]
 - The probability that a 25-year-old lives die between ages 40 and 45. [1 mark]
- (d) i. An insurance company has two types of policy, A and B. Claims arriving under A follow a Poisson process with a rate of 5 per day. Claims arrive independently under B and follow a Poisson process with a rate of 3 per day. A randomly selected claim from B has a probability of two-thirds of being over 10000 dollars. How many claims over 10000 dollars would you expect tomorrow. [3 marks]
- ii. Define an n -year deferred annuity. [1 mark]
- (e) i. Sophy is 30 years. Beginning one year from today, she will be receive 20000 dollars annually for as long as she is alive. Obtain an expression for the present value of this series of payments assuming an interest rate of 9 percent. [2 marks]
- ii. Give a formular for a temporary life annuity which will only continue for a maximum of n years. [2 marks]
- (f) Given $l_x = 1000(1 - \frac{x}{105})$, determine each of the following expressions;
- ${}_{15} q_{25}$
 - The probability that a 30-year-old dies between ages 55 and 60



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 - The probability that a 25-year-old lives die between ages 40 and 45. [1 mark]
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- ${}_{15}q_{25}$
 - The probability that a 30-year-old dies between ages 55 and 60

- iii. The probability that a 30-year-old dies after age 70.
- iv. The probability that, given a 20-year-old and a 30-year-old, one but not both of these individuals reaches age 70.
- v. The probability that a 15-year-old reaches age 110. [5 marks]

QUESTION TWO

- (a) i. A researcher studies the mortality patterns of rats. He establishes the following probabilities of deaths; $q_0 = 0.4$, $q_1 = 0.20$, $q_2 = 0.30$, $q_3 = 0.70$, $q_4 = 1$. Starting with l_0 , construct a life table with functions l_x , d_x and q_x . [6 marks]
- ii. Reported claims from a certain group of policies follow a Poisson process with a uniform rate of 5 per day (and the insurer has a 24 hour hotline). Compute;
 - A. the probability that there will be fewer than two claims reported on a given day. [2 marks]
 - B. the probability that another claim will be reported during the next hour. [2 marks]
- (b) i. Give and explain different types of annuities in life contingencies. [3 marks]
- ii. Find the net single premium for a life annuity of 500 per year, with the first payment due in one year, sold to a 30-year-old in each of the following cases;
 - A. $p_x = 0.96$ for each x and $i = 0.09$.
 - B. $l_x = 1000(1 - \frac{x}{115})$, and $i = 0.13$. [6 marks]
- iii. Define the concept commutation functions. [1 mark]

QUESTION THREE

- (a) i. Evanson Odiwuor, aged 40, purchases a whole life policy of 50000. If $N_{40} = 5000$, $N_{41} = 4500$ and $i = 0.08$, find the price of this policy. [5 marks]
- ii. Find the net single premium for 100000 life insurance policy payable in 20-year period, purchased by a life aged 30 if it is assumed that $i = 0.06$ and ${}_t p_{30} = 0.98^t$ for all t . [5 marks]
- iii. Explain the concept pure endowment and give the value of this payment at present. [2 marks]
- (b) i. Use the extract below from a life table to solve the following questions;

Age	l_x	d_x	$1000q_x$
0	1000000	1580	1.58
1	998420	680	.68
2	997740	485	.49
3	997255	435	.44

- A. ${}_4|_3p_1$
- B. The probability that the newborn will survive the first year, but die in the first two months thereafter. [4 marks]

- ii. There are four persons now aged 40, 50, 60 and 70. Find an expression for the probability that the 40-year-old and the 50-year-old will die within the five year period and neither the 60-year-old nor the 70-year-old will die during the five year period. [3 marks]

QUESTION FOUR

- (a) i. Give a special probability that lives (x) and (y) die within n years by the common shock. [2 marks]
 ii. Show that for two lives aged (x) and (y) ,

$${}_t p_{xy} = {}_t p_x \cdot {}_t p_y \cdot e^{-\lambda t}.$$

[4 marks]

- iii. Explain the concepts
 A. Last survivor status model
 B. Common shock model.
- (b) A mortality table has a select period of three years. Find expressions in terms of the life table functions $l_{(x)+t}$ and l_y for
- q_{50} [2 marks]
 - ${}_2 p_{50}$ [2 marks]
 - ${}_2 | q_{50}$ [3 marks]
 - ${}_2 | 3 q_{[50]+1}$ [3 marks]

QUESTION FIVE

- (a) i. Given $l_{xy} = 1000$, $l_{x+10:y} = 960$ and $l_{x:y+10} = 920$, calculate the probability that of two lives aged (x) and (y) , only one will survive for 10 years. [5 marks]
 ii. A researcher asks: Is it appropriate to assume that transition intensities are constant? Surely, the chance of becoming sick or dying increases with age. Comment on this question. [5 marks]
- (b) i. During a large study into rates of sickness the following occurred;
 2710 healthy lives fell sick and 2490 sick lives recovered.
 70 healthy lives and 120 sick lives died.
 For the whole group, the periods of health and sickness totalled 41200 and 6700 years respectively. Estimate the annual forces of transition between the states of healthy, sick and dead. [5 marks]
 ii. Give the assumptions underlying the simple two-state model. [5 marks]

- iii. The probability that a 30-year-old dies after age 70.
- iv. The probability that, given a 20-year-old and a 30-year-old, one but not both of these individuals reaches age 70.
- v. The probability that a 15-year-old reaches age 110. [5 marks]

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MAIN CAMPUS

MAC 306: FINANCIAL ECONOMICS

Date: 12th July, 2018

Time: 3.30 - 6.30pm

INSTRUCTIONS:

- Answer Question ONE and any other TWO.



QUESTION 1 [COMPULSORY] [30 Marks]

(a) State the expected utility theorem. [2 Marks]

(b) By considering the relationship $R(w) = w.A(w)$, explain which of the following statements is true.

1. If an investors preferences display decreasing relative risk aversion then they must also display decreasing absolute risk aversion.
2. If an investors preferences display decreasing absolute risk aversion then they must also display decreasing relative risk aversion.

[4 Mark]

(c) A horse trading fan assesses her utility of wealth using the utility function;

$$U(w) = 2(w^{0.5} - 1)$$

Prove algebraically that the horse trading fan is;

(i) Non-satiated [4 Marks]

(ii) Risk averse [4 Marks]

(d) Consider two risky assets A and B with the cumulative distribution function

$$F_A(w) = w$$

$$F_B(w) = w^{\frac{1}{2}}$$

In both cases $0 \leq w \leq 1$.

Show that A is preferred to B on the basis of first order stochastic dominance. [8 Marks]

(e) Consider assets A and B which offer returns as set out in the following table.

Asset	Return, $\bar{r}$ (%)	Standard Deviation, σ (%)
A	10	15
B	18	30

The correlation coefficient between the assets is 0.4

- (i) Find the proportion α of A and $(1 - \alpha)$ of B that define a portfolio of A and B having a minimum standard deviation. [4 Marks]
- (ii) What is the value of this minimum standard deviation?. [2 Marks]
- (iii) What is the expected return of this portfolio? [2 Marks]

QUESTION 2 [20 Marks]

- (a) Define first and second-order stochastic dominance. [5 Marks]
- (b) Consider four assets which deliver a one year return r_i on asset i with the probabilities as set out below:

	$P(r_i = -5\%)$	$P(r_i = -3\%)$	$P(r_i = 0\%)$	$P(r_i = 3\%)$	$P(r_i = 5\%)$
Asset 1	0.2	0.2	0.2	0.2	0.2
Asset 2	0.3	0.2	0.1	0.2	0.2
Asset 3	0.1	0.3	0.2	0.3	0.1

Determine which dominance, if any, is exerted by;

- (i) Asset 2 over asset 3 [5 Marks]
- (ii) Asset 3 over asset 1 [5 Marks]
- (iii) Asset 1 over asset 2 [5 Marks]

QUESTION 3 [20 Marks]

A risk averse investor makes decisions using a quadratic utility function

$$U(w) = w + dw^2$$

- (a) Derive an upper bound for d for this investor. [5 Marks]
- (b) Explain why the investor can only use this utility function to make decisions over a limited range of wealth, w . Your answer should include a statement of this range. [5 Marks]
- (c) The investor states that the upper limit of wealth where she can use this utility function is $w = Kshs.1,000$. Determine the value d in the investor's utility function. [2 Marks]
- (d) The investor wins a price of Kshs.250 in a game show. She is then offered the opportunity to exchange this price for a larger price of Kshs.600 if she can answer one more question. However she will receive no price at all if she gets the question wrong. She estimates her chances of answering the question correctly to be 50%. Determine whether the investor should take this opportunity to exchange. [8 Marks]

QUESTION 4 [20 Marks]

In a market in which the Capital Asset Pricing Model(CAPM) holds, there are two securities with the following attributes (expressed per annum).

	Security	A	B
$E(r_i)$		0.196	0.164
$Cov(r_i, r_j)$	A	0.05	0.01
	B	0.01	0.03

- (a) Determine the composition of the market portfolio with expected return of 18% per annum. [6 Marks]
- (b) Calculate the beta of each security, under the assumption that the risk-free rate of interest is 10% per annum. [6 Marks]
- (c) State the limitations of the CAPM. [8 Marks]

QUESTION 5 [20 Marks]

- (a) Define in the context of the mean-variance portfolio theory
- (i) an inefficient portfolio [2 Marks]
- (ii) an efficient portfolio [2 Marks]
- (b) An investment universe includes two assets, A and B , with expected return on asset i of r_i and variance v_i as set out below:

Asset i	Expected return (r_i)	Variance (v_i)
A	$r_A = 0.05$	$v_A = 0.16$
B	$r_B = 0.07$	$v_B = 0.25$

The correlation of returns $c_{AB} = -0.2$. In an efficient portfolio, let a be the proportion which is held in asset A.

(i) Express the portfolio variance in terms of a quadratic function in a , showing your workings. [5 Marks]

(ii) Let R be the expected return on the portfolio. Express the portfolio variance in terms of a quadratic function in R , using your result in b(i) and showing your workings (Your expression should not include a).

[5 Marks]

(iii) The expression in (ii) above represents the efficient frontier. An investor uses the utility function that gives rise to an indifference curve

$$V = 16R - 200R^2$$

Determine the two portfolios on the efficient frontier that also lie on the investor's indifference curve. [6 Marks]